positions. However, Tissot and Blaise reported the Sm-type structure for an alloy of 29.5 at.% Pr, but found alloys of 19.6 and 36.2 at.% Pr to be too complex to index.

Metastable Solid Solutions

Wang [76Wan] splat cooled Gd-Pr alloys that had compositions normally in the cph phase field. By splat cooling, the cph phase field was extended into the composition range that under equilibrium conditions would have the Sm-type structure. The Sm-type structure was not suppressed, but the cph phase was extended to higher Pr concentrations. Wang found that the extension of the cph phase field was limited by a c/a ratio of ~1.597.

Thermodynamic Properties

Dreyfus *et al.* [63Dre] measured the heat capacity of a Gd–27 at.% Pr alloy from 1.4 to 4 K. Their data were fitted to the following equation:

$$C = AT^{-2} + \gamma T + \beta T^3$$

where the first term is the nuclear contribution, the second term is the electronic contribution, and the third term is the lattice contribution to the heat capacity. The values obtained for the constants are: $A = 228 \pm 2.2$ mJ-K/mol, $\gamma = 26.3 \pm 0.6$ mJ/mol-K^2 and $\beta = 1.16 \pm 0.06$ mJ/mol-K^4.

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Gd-Pr evaluation contributed by **K.A. Gschneidner, Jr.**, Director, and **F.W. Calderwood**, Rare-earth Information Center, Ames Laboratory, Iowa State University, Ames, Iowa 50011. This work was supported by the Department of Energy through the Joint Program on Critical Compilation of Physical and Chemical Data coordinated through the Office of Standard Reference Data, National Bureau of Standards. Additional support was contributed by: Th. Goldschmidt AG, Essen, West Germany; Molycorp, Inc., Union Oil Co. of California, Los Angeles, CA; Reactive Metals & Alloys Corp., West Pittsburg, PA; Ronson Metals Corp., Newark, NJ; and Santoku Metal Industry Co. Ltd., Kobe, Japan. Literature searched through 1981. Professor Gschneidner is the ASM/NBS Data Program Category Editor for binary rare-earth alloys.

The Nd-Pr (Neodymium-Praseodymium) System

144.24 140.9077

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Phase Diagram

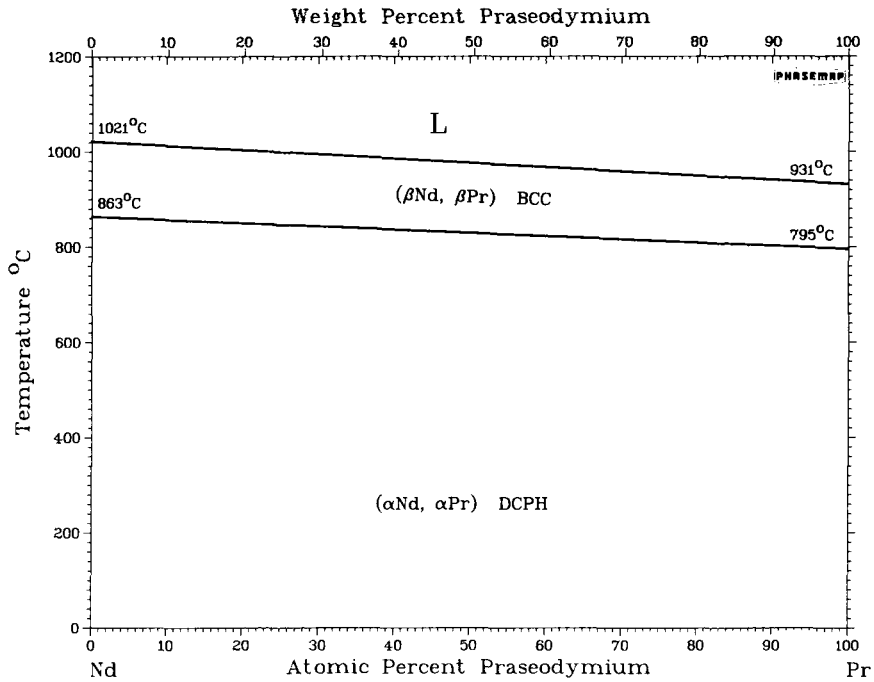
The phase diagram for the Nd-Pr system was investigated by Markova *et al.* [63Mar] and by Lundin *et al.* [64Lun]. Markova *et al.* used 97 (wt.?)% pure Pr and 98% pure Nd. The impurities present in their metals consisted of other rare-earth metals, plus Cu, Fe, and Ca in unspecified amounts. Their methods of investigation included thermal analysis, metallography, X-ray diffraction, and hardness measurements. Their alloys were annealed at 600 °C for 20 h before measurements were made. Terekhova [63Ter] published a similar phase diagram that appeared to be based on the same research. Each diagram shows complete solid solubility for both the dcph and the bcc structures and a definite two-phase region (liquid plus solid) between the liquidus and solidus lines.

Lundin *et al.* [64Lun] determined the phase diagram for this system using metallography, thermal analysis, X-ray diffraction, and density measurements. The total content of other rare earths in their starting metals did not exceed 0.16 wt.%. Impurities reported in their Pr were: 65 ppm C, 50 ppm N, 27 ppm F, <0.02% Ca, <0.025% Si, <0.01% Mg, and ~0.05% Ta. Impurities in their Nd included: 62 ppm C, 39 ppm N, 200 ppm F, 0.01% Fe, <0.03% Ca, <0.025% Si, <0.01% Mg, and 0.05% Ta. Alloys were arc melted under purified argon and remelted four times to improve the homogeneity. Care was taken in the preparation of metallographic specimens to avoid reaction products and disturbed metal. Screened filings for X-ray diffraction studies were annealed to relieve cold working. Densities were determined by displacement in monobromobenzene as well as by calculation from lattice parameters.

The phase diagram for the Nd-Pr system is shown in Fig. 1. The equilibrium diagram as presented by Lundin *et al.* showed narrow two-phase regions between the liquidus and solidus lines and between the $\alpha \rightarrow \beta$ transformation lines. Their diagram showed complete solubility for both the dcph and the bcc allotropes. Because Nd and Pr are both trivalent, have adjacent atomic numbers, and exhibit a continuous series of solid solutions, the liquidus and solidus lines in Fig. 1 were drawn as a single straight line connecting the two melting points of Nd and Pr (see [82Gsc]). For the same reason, the upper and lower solvus curves are also drawn as a single straight line. In addition, the melting point temperatures and the transformation temperatures of the pure metals have been adjusted to the accepted values listed in [78Bea].

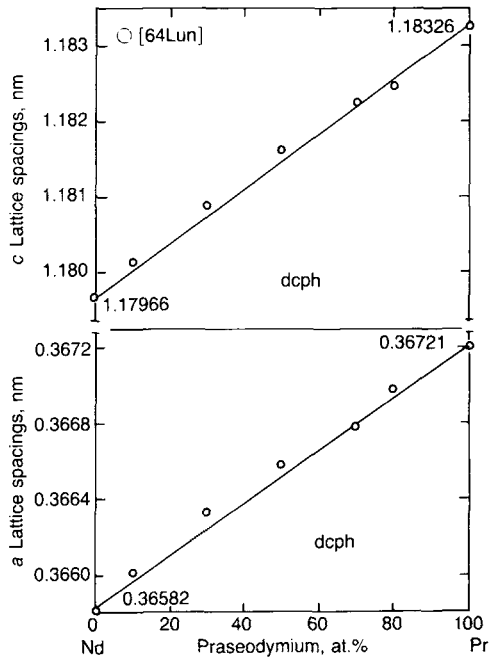
In addition to the experimental determinations of the phase diagrams described above, Shiflet *et al.* [79Shi] calculated the phase diagram for the Nd-Pr system using

Fig. 1 Nd-Pr Phase Diagram



K. A. Gschneidner, Jr. and F. W. Calderwood, 1982.

Fig. 2 Lattice Spacings in the Nd-Pr System



Straight lines represent the Vegard's law relationship based on the accepted data for the pure metals, as shown in [78Bea].
K. A. Gschneidner, Jr. and F. W. Calderwood, 1982.

the differences in enthalpy and entropy between the different crystal structures in the pure elements and the regular solution constant for the enthalpy of mixing. The paths of two-phase fields that Shiflet *et al.* calculated for this diagram were in good agreement with those determined experimentally by Markova *et al.* and by Lundin *et al.*, but the widths of those fields were appreciably smaller than those experimentally observed. These wide two-phase fields would be expected to occur because of impurities present in metals used in the experimental research (see [82Gsc]).

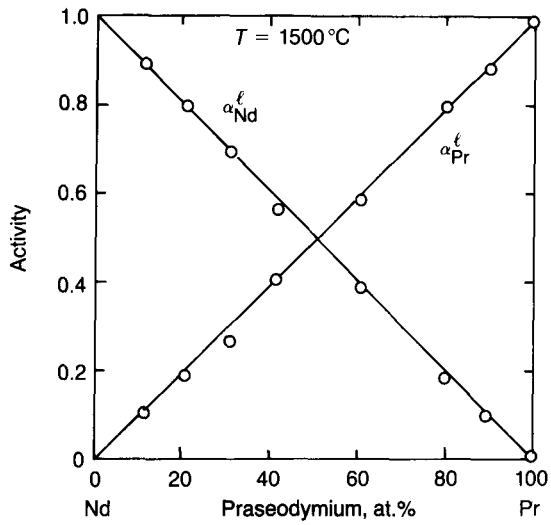
Lattice Spacings

Lundin *et al.* [64Lun] determined lattice spacings of their pure metals and of a series of alloys. Because the *a* and the *c* spacings of their pure metals differ slightly from the accepted values for these spacings (see [78Bea]), adjustments have been made in the spacing data of the alloys on the basis of the observed differences in the spacings for the pure metals and the mole fraction of each component in the alloy. These adjusted values for the *a* and *c* lattice spacings are plotted in Fig. 2. A straight line connecting the *a* spacing of pure Nd with the *a* spacing of pure Pr, and a similar straight line connecting the *c* spacings of these pure components, indicates the Vegard's law relationship in this system. These investigators also measured density of the various alloys by displacement in monobromobenzene and found excellent agreement with densities determined by calculations using their lattice spacings. Their density vs composition plot was linear, another evidence of ideality that is unusual in binary metal systems.

Thermodynamic Properties

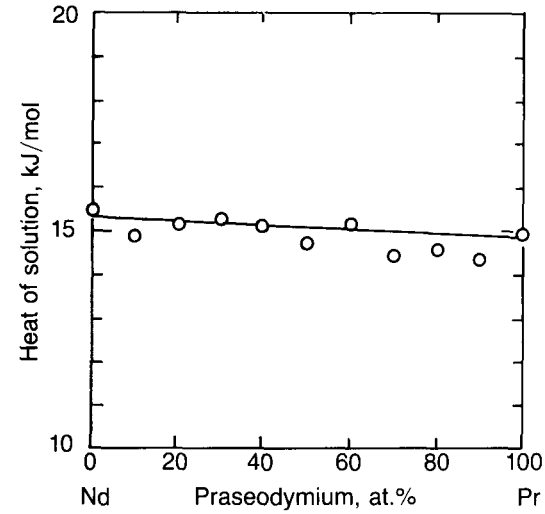
Lundin *et al.* [65Lun1] measured the thermodynamic activities at 1475, 1500, and 1525 °C of liquid solutions of Nd-Pr. Vapors from eight alloys were condensed and the

Fig. 3 Activities of Nd and Pr in Liquid Nd-Pr Alloys at 1500 °C



Essentially identical results were obtained at 1475 and 1525 °C.
K. A. Gschneidner, Jr. and F. W. Calderwood, 1982.

Fig. 4 Heat of Solution in Liquid Sn at 477 °C for Nd-Pr Alloys



As measured by liquid metal solution calorimetry.
K. A. Gschneidner, Jr. and F. W. Calderwood, 1982.

compositions were determined by X-ray spectrographic analysis; from these data, the activity of each component was calculated. Figure 3 is a plot of the activity of liquid Nd and liquid Pr as a function of the mole fraction of Pr in the liquid state at 1500 °C. The activities vs composition at the other two temperatures are similar to the results obtained at 1500 °C. The linear relationship observed in the plot indicates that the solutions are essentially ideal because the activity is equal to the mole fraction. Lundin *et al.* [65Lun2] measured the heats of solution in liquid Sn at 477 °C of pure Pr, pure Nd, and nine Nd-Pr binary alloys using liquid metal calorimetry. Figure 4 is a plot of the measured heat effect for the alloys as a function of composition. A least-squares analysis of their data produced a straight line fit. Because a linear relationship exists, the heat of mixing in the binary Nd-Pr system is zero within experimental error.

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Nd-Pr evaluation contributed by **K. A. Gschneidner, Jr.,** Director, and **F. W. Calderwood,** Rare-earth Information Center, Ames Laboratory, Iowa State University, Ames, Iowa 50011. This work was supported by the Department of Energy through the Joint Program on Critical Compilation of Physical and Chemical Data coordinated through the Office of Standard Reference Data, National Bureau of Standards. Additional support was contributed by: Th. Goldschmidt AG, Essen, West Germany; Molycorp, Inc., Union Oil Co. of California, Los Angeles, CA; Reactive Metals & Alloys Corp., West Pittsburg, PA; Ronson Metals Corp., Newark, NJ; and Santoku Metal Industry Co. Ltd., Kobe, Japan. Literature searched through 1981. Professor Gschneidner is the ASM/NBS Data Program Category Editor for binary rare-earth alloys.